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Next day discharge after the Nuss procedure using intercostal nerve cryoablation, intercostal nerve blocks, and a perioperative ERAS pain protocol

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ABSTRACT

Background: The Nuss procedure for pectus excavatum has historically been associated with significant postoperative pain, which has been the major factor contributing to hospital length of stay (LOS).
Methods: A single-institution, prospective study of 40 consecutive patients undergoing Nuss bar placement for pectus excavatum between November 2019 and January 2021 was conducted to assess the effectiveness of a multimodality pain management protocol. All patients received T3-T8 intercostal nerve cryoablation (INC), T3-T8 bupivacaine intercostal nerve blocks, Exparel at the skin incisions, and management with a perioperative analgesia regimen that minimized narcotic usage. The primary outcome was LOS. Secondary outcomes included opioid use, pain scores, and time to sensory recovery.
Results: 37/40 patients (92.5%) were discharged home on postoperative day (POD) 1, and 3/40 (7.5%) were discharged on POD 2 (mean LOS = 1.1 days). The median average postoperative pain score was 2/10. After eliminating IVPCA from our protocol, total oral morphine equivalent (OME) decreased by 73% (55.5 mg to 15 mg) with no change in pain scores or discharge timing.
Conclusions: INC combined with bupivacaine intercostal nerve blocks and a pre- and post-hospital analgesia protocol facilitated discharge one day after the Nuss procedure, achieved excellent pain control, and eliminated the need for intravenous opioids.

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1. Introduction

Pectus excavatum affects approximately 1 in 1000 live births and represents 90% of chest wall deformities [1,2]. The Nuss procedure is the most common operation used to repair pectus excavatum. While it is minimally invasive, the process of placing a bar posterior to the sternum to elevate and remodel the chest is associated with significant postoperative pain.

Pain control remains the single most important factor contributing to hospital length of stay (LOS) following the Nuss procedure. Moreover, inadequate pain control can reduce patient satisfaction, limit mobility, increase opioid consumption, and increase healthcare visits and re-admissions following discharge. For these reasons, optimisation of postoperative pain is an integral part of caring for patients undergoing a Nuss procedure. Historically, thoracic epidural was the mainstay of postoperative analgesia [3], but it was often limited by unreliable pain control, significant length of stay, and in rare cases substantial complications. Many groups, including our own, moved away from thoracic epidurals towards patient controlled analgesia (PCA) after several studies, including randomised controlled trials, showed no clear superiority between the two [4,5]. More recently, the addition of intercostal bupivacaine nerve blocks showed improvement in pain scores and LOS [6,7]. Video assisted intercostal nerve cryoablation (INC) is the newest method that surgeons are adding to their armamentarium during the Nuss procedure [8–16].

Intercostal nerve cryoablation has been used for decades, most notably for pain control in patients who have undergone a thoracotomy, where it proved effective in decreasing postoperative opioid usage and improving pulmonary mechanics [17,18]. More recently, it has been used in thoracoscopic lung procedures and in conjunction with the plating of rib fractures. Since the Food and Drug Administration (FDA) approved the first commercially available cryoablation probe in 2015, there has been increasing interest in adopting cryoablation for use during the Nuss procedure. Modern cryoprobes temporarily prevent pain transmission

Abbreviations: DOS, day of surgery; ERAS, enhanced recovery after surgery; INC, intercostal nerve cryoablation; IVPCA, intravenous patient controlled analgesia; LOS, length of stay; OME, oral morphine equivalent; PDD, post discharge day; POD, postoperative day.

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Table 1
Perioperative analgesia regimen.

	4 Days Before Surgery	2 Days Before Surgery	DOS	POD 1 (Anticipated Discharge)	PDD 1,2	PDD 3–7	PDD 8–10
Gabapentin	300 mg PO tid	300 mg PO tid	400 mg PO tid	400 mg PO tid	400 mg PO tid	400 mg PO tid	400 mg PO tid
Celecoxib	OFF	200 mg PO bid	200 mg PO bid	200 mg PO bid	200 mg PO bid	200 mg PO bid	OFF
Acetaminophen	OFF	650 mg PO qid	650 mg IV Q6 hrs*	650 mg PO Q6 hrs	650 mg PO qid	650 mg PO qid	650 mg PO qid
Diazepam	OFF	OFF	2 mg IV Q12 hrs*	2 mg PO bid	2 mg PO bid	OFF	OFF
Oxycodone	OFF	OFF	5–10 mg PO Q6h PRN	5–10 mg PO Q6h PRN	5–10 mg PO Q6h PRN	OFF	OFF
Ondansetron	OFF	OFF	4 mg IV Q6 hrs*	4 mg PO tid PRN	4 mg PO bid PRN	OFF	OFF

DOS = Day of Surgery, POD = Postoperative Day, PDD = Post Discharge Day.
* Acetaminophen, diazepam and ondansetron were changed from IV to PO for the last 5 patients.

via rapid, localised freezing of nerves, which induces disruption of axons and demyelination (axonotmesis or a grade II nerve injury) [17]. This process leaves the fibrous neural structures including the epineurium, perineurium and endoneurium intact, and the process of Wallerian degeneration eventually results in axonal regeneration and reinnervation of the chest wall. Since nerve fibers can grow into their original endoneurial tubes, the original fiber pattern is maintained.

Interpretation of the current literature reporting the experience with cryoablation during the Nuss procedure is limited by small sample sizes, incomplete reporting of pain scores, opioid consumption and sensory recovery, and lack of uniformity in analgesic approaches. This prospective study details our experience using a comprehensive enhanced recovery after surgery (ERAS) pain pathway for patients undergoing the Nuss procedure in one of the largest and most complete series to date.

2. Methods

2.1. Study population and design

We conducted a single-institution, prospective review of consecutive patients undergoing the Nuss procedure for correction of pectus excavatum between November 2019 and January 2021. There were two primary surgeons for these patients: J.W.D. (n = 37) and A.L.D. (n = 3). No patients were excluded. The primary outcome was length of stay, and secondary outcomes included postoperative opioid use, pain scores, and time to full sensory recovery. Postoperative opioids were defined as total intravenous and oral opioids converted to oral morphine equivalents (OMEs) to allow for comparison of multiple medication types.

2.2. Peri-operative enhanced recovery after surgery (ERAS) medication protocol

Our perioperative medical analgesia regimen is detailed in Table 1. Briefly, all patients received gabapentin starting 4 days before surgery and acetaminophen and celecoxib starting 2 days before surgery. These three medications were also taken the morning of surgery and continued post operatively. Diazepam and as needed oxycodone were added immediately after surgery. There has been some evolution of the protocol during the course of this study. Initially, on the night of surgery, acetaminophen was given intravenously and celecoxib was replaced with IV ketorolac until the morning of postoperative day (POD) number 1, at which time acetaminophen was changed to the oral route and ketorolac was replaced with oral celecoxib. In the last 5 patients of the series, we stopped using IV acetaminophen and IV ketorolac and now simply continue the oral acetaminophen and celecoxib. Moreover, in the first 19/40 patients we used demand only hydromorphone (Dilaudid) IVPCA on the night of surgery, but for the most recent 21/40 patients in the series, no IVPCA was used. Therefore, our current protocol includes only oral medication with no intravenous medications whatsoever. At discharge, all patients received a two day prescription for eight 5 mg tablets of oxycodone, and they continued diazepam for 2 days, celecoxib for 7 days and gabapentin and acetaminophen for 10 days.

2.3. Intraoperative technique: cryoablation

Cryoablation and intercostal nerve blocks were uniformly performed prior to Nuss bar placement to reduce intraoperative anesthetic requirements. A double lumen endotracheal tube was used in all cases to provide selective lung ventilation. Starting on the right side, a 4 cm incision was made in the axilla along the line for bar placement. The lung was deflated and the chest insufflated to 8 cm H_20 using a veress needle placed through the incision. A 5 mm port was then placed through the axillary incision into the right chest, and a 5 mm 30 ° scope was used to identify a suitable interspace for placement of a second 5 mm port just above the diaphragm in the anterior axillary line. The camera was moved to this port. The table was rotated left side down to better expose the intercostal nerves. The first 5 mm port was removed, and its entry site was widened slightly to allow introduction of the Atricure CryoICE probe (Mason, OH) into the chest directly through the axillary incision without a port Fig. 1.

Sites for intercostal nerve cryoablation were identified 4 cm anterior to the edge of the vertebral column for intercostal nerves T3-T8. The tip of the cryoprobe was placed sequentially in each intercostal groove. For each nerve, a 2 min freeze cycle was applied at a temperature of -67 °C followed by a 30–45 s thaw cycle, after which the probe was safely removed without applying traction to the nerve. Care was taken to avoid contact between any part of the cryoprobe and the lung. Cryoablation was then performed in similar fashion on the left side.

In some patients with more severe indentations of the chest, the heart was shifted dramatically to the left and was found in contact with the left chest wall. In these instances (identified on axial imaging), the first 5 mm port on the left side was placed several centimeters cephalad to the axillary incision (instead of through it) to avoid cardiac injury. After establishing the pneumothorax through this port, the camera was then inserted to guide safe placement of the inferior camera port near the diaphragm.

2.4. Intraoperative technique: bupivacaine intercostal nerve blocks

After cryoablation was completed, 0.25% bupivacaine with 1:200,000 epinephrine intercostal nerve blocks were placed immediately anterior to the cryoablation sites from T3-T8. The nerve

Fig. 1. The AtriCure CryoSPHERE ©.

blocks were placed using a long 25 gaugeneedle to inject just beneath the pleura. Prior to injection, aspiration was performed to prevent intravascular injection. When aspiration revealed blood (4 occurrences in 480 total nerve blocks) the needle was removed and pressure was held with a laparoscopic peanut, which provided hemostasis in all cases and after which the nerve block was still performed. The bupivacaine visibly spread proximal and distal to the cryoablation site. A maximum of 1 mL/kg of bupivacaine may be used, so for 12 nerve blocks in a 60 kg patient (T3-T8 bilaterally) the maximum dose is 5 mL per nerve block. The maximum dose of bupivacaine was used in all cases. The addition of cryoablation and intercostal nerve blocks added approximately 80−90 min to the operative time.

2.5. Intraoperative technique: bar placement

Sternal elevation was performed in all cases using a tenaculum to grasp the sternum and attaching it to the Ruhltract sternal elevation retractor (Cleveland, OH). Each bar was secured with one stabiliser. In cases using one bar, the stabiliser was placed on the left by convention and multiple 0 PDS ties were placed on the right side around the bar and corresponding rib. In cases using two bars, one stabiliser was used for each bar and the 0 PDS ties were not used. In the single case where three bars were used, each bar had one stabiliser and no PDS ties.

2.6. Intraoperative technique: closure

The maximum allowable dose of 20 mL bupivacaine liposome injectable suspension (Exparel) was used to provide 48–72 h of local anesthesia on the axillary skin incisions. If there was one axillary incision on each side (i.e. for placement of one bar), 10 mL was used for each incision. If two bars were used, separate axillary incisions were made for each bar, so the 20 mL Exparel was diluted 1:1 with normal saline, resulting in 40 mL that was divided equally between the 4 axillary incisions. Foley catheters were removed in the operating room.

2.7. Follow-up

Following discharge, requests for refills of oxycodone were tracked as a gaugeof need for additional postoperative narcotics. Patients were subsequently seen in clinic at one, three, six and twelve months after surgery. Patients were examined for sensory recovery at each visit and monitored for chronic neuropathic and non-neuropathic pain.

Table 2
Patient characteristics ($n = 40$).

Sex (male)	80% (32/40)
Age (years)	15 (3, 13, 29)[a]
Weight (kg)	56 (19.5, 36.7, 95.6) [a]
Overall Haller Index	4.7 (1.8, 3.1, 9.7)[a]
Haller Index, Patients <18 Years Old	4.7 (1.9, 3.1, 9.7) [a]
Haller Index, Patients ≥ 18 Years Old	5.0 (1.8, 3.2, 6) [a]
One Bar Placed	42.5% (17/40)
Two or More Bars Placed	57.5% (23/40)
Operative Time/ 1 Bar (min)	152.5 ± 17.4[b]
Operative Time/ 2 or More Bars (min)	200.9 ± 35.28[b]

[a] Data are presented as median (IQR, minimum, maximum).
[b] Data are presented as mean ± SD.

2.8. Statistical analysis

Statistical analyses were performed with GraphPad Prism 9.1.0 (San Diego, CA). Continuous variables were tested for normality of distribution using the Kolmogorov-Smirnov test. Summary statistics for continuous variables are primarily presented as median (interquartile range [IQR], minimum, maximum) due to the predominance of nonparametric data. Select data are presented as mean ± standard deviation when normally distributed. Categorical variables are presented as frequency and percentage. The Mann-Whitney U test was used to compare two groups of non-parametric data, and a p-value < 0.05 was considered significant.

3. Results

3.1. Patient characteristics

Demographics and operative characteristics are detailed in Table 2. The majority of patients in this study were <18 years old; however, we did include 8 adult patients (defined as 18 years and older) in our study. These patients had a median Haller index of 5.0, as well increased chest wall stiffness by virtue of their age/skeletal maturity. 23/40 (57.5%) of patients required more than one bar, with one patient requiring 3 bars.

Table 3
Pain control and postoperative opioid requirements[a].

	Overall(n = 40)	With PCA in Protocol (n = 19)	Without PCA in Protocol (n = 21)	p-value[b]
Average Postoperative Pain Score*	2 (1.9, 0, 5.1)	2 (2, 0, 3)	2 (2.3, 1, 5.1)	0.49
Total Postoperative Opioid OME (mg)[c]	30 (40.9, 0, 182)	55.5 (77.5, 10, 182)	15 (33.8, 0, 52.5)	0.0001
Postoperative IV PCA OME (mg)	N/A	48 (76, 8, 168)	N/A	N/A
Postoperative Oxycodone OME (mg)	7.5 (7.5, 0, 37.5)	7.5 (15, 0, 37.5)	7.5 (11.3, 0, 37.5)	0.2260
Equivalent Number of Oxycodone 5 mg Tablets for Total Postoperative OME	4.0 (5.5, 0, 24.4)	7.4 (10.3, 1.3, 24.4)	2.0 (4.5, 0, 7.0)	0.0001

OME = oral morphine equivalent.
* Pain score is out a of a total of 10.
[a] Data are presented as median (IQR, minimum, maximum).
[b] Comparing data with and without a PCA in the protocol.
[c] Includes postoperative oxycodone OME, IV PCA OME (when given) and OME for any additional narcotics given (e.g., a small dose of fentanyl in the PACU).

3.2. Length of stay

The vast majority of patients in this study (37/40, 92.5%) were discharged home the day after surgery, and 3/40 (7.5%) were discharged on POD number 2 for an average LOS of 1.1 ± 0.27 days. Two patients who stayed in the hospital an extra day did so because of pain. One of these patients was a 26 year old male, our very first cryoablation patient, and the other was a 14 year old female. One patient was ready for discharge on POD number 1 but stayed an extra day because she did not have transportation home. The two patients staying an extra day due to pain were early in the series, and all of our last 27 patients were successfully discharged on POD number 1. Excluding the one 26 year old patient mentioned above, all of our other adult patients (18 years and older) went home on POD number 1 with no increase in pain scores or postoperative opioid requirements compared to their younger counterparts.

3.3. Pain control and opioid usage

The median average postoperative pain score was 2/10. The postoperative opioid requirements to achieve this level of pain control were low and are summarized in Table 3. Briefly, patients managed on the protocol that included an IVPCA used a total postoperative oral morphine equivalent (OME) of 55.5 mg (which equates to 7.4 tablets of 5 mg oxycodone), whereas those managed without an IVPCA used a total postoperative OME of only 15 mg (which equates to 2 tablets of 5 mg oxycodone). Therefore, simply eliminating the IVPCA reduced the total postoperative OME by 73%. With this change, there were no differences noted in pain scores, discharge timing, or oxycodone usage between cohorts. Similarly, the last 5 patients in our series that were managed without IV acetaminophen or IV ketorolac had no increase in pain scores. 8/40 patients (20%) took no oxycodone at all, including 2 patients that were managed without a PCA. Only 3/40 patients (7.5%) requested a refill of their 8 tablet prescription following discharge, and no patients were readmitted for pain control.

Average pain score for the 22 patients with two bars was 2.4 and for the 17 patients with one bar was 2.1. The single patient with three bars had an average pain score of 1.5. These minor differences in pain scores between patients with one bar versus two bars were not statistically significant or clinically meaningful.

3.4. Sensory recovery

By three months after surgery 7/25 patients (28%) had normal chest wall sensation and by six months after surgery 17/20 (85%) had normal chest wall sensation (subjectively and by physical exam). Of the 3/20 patients still noticing some numbness at 6 months, 2 patients had a 2 cm band of numbness across the chest just below the nipple line and one patient had a small 2 × 2 cm area of numbness on the central chest over the mid-sternum. In the first three months, 10% (4/40) of patients described their chest wall as “hypersensitive,” and hyperesthesia to light touch was confirmed on physical exam. This resolved in all cases. 2/40 patients described transient “tingling sensations” which also resolved. With an average follow up of 12 months (range 6–20 months) there were no patients who experienced chronic neuropathic pain. All of the 20 patients who were at least one year out from their surgery had normal chest wall sensation, with no areas of residual numbness by 12 months.

4. Discussion

The minimally invasive Nuss procedure has still been associated with significant postoperative pain. The newest option for postoperative pain control is intercostal nerve cryoablation, and it has shown promise in accelerating postoperative recovery. Several groups have decreased LOS to between 2 and 4 days using this modality [8–11,14]. Five studies, including one small randomised controlled trial, have specifically compared outcomes using thoracoscopic INC to those achieved with thoracic epidural [8–11,14]. These studies consistently demonstrated reductions in average hospital length of stay from 5 to 6 days with epidurals to 2,3 days with cryoablation. Still, several of the patients in these series required longer hospitalisations. The largest reported experience included 40 patients, and while the average length of stay was 2.5 days, 35% of patients were discharged ≥ POD number 3 [14]. Moreover, another relatively large experience of 43 patients undergoing cryoablation demonstrated that even with cryoablation, hospital length of stay was prolonged in patients >21 years old that presumably had less compliant chest walls (3.9 days vs. 2.0 days for those <21 years old, median age 16) [16].

A more recent report has shown further decreases in LOS using cryoablation alone. Dekonenko et al. compared 35 prospectively collected patients receiving cryoablation with data from a recently completed RCT of 32 epidural and 33 PCA patients [4]. They noted a decrease in LOS to a median of 1 day in the cryoablation group compared to means of 4.3 and 4.2 days in the epidural and PCA groups, respectively [4]. While these results are on par with ours in terms of length of stay, they did have some patients in the hospital for up to 3 days postoperatively, and they do not report how many patients stayed longer than 1 day.

Our series of 40 patients remains among the largest published, and our outcomes are the most consistent for a series of that size. 37/40 patients (92.5%) were discharged home the day of surgery and 3/40 (7.5%) were discharged on POD number 2 for an overall LOS of 1.1 days. The three patients that required more than one night in the hospital were among our early experience. Of these three, one stayed the extra day strictly for social reasons, and one was a 26 year old male (our first cryoablation patient) who was therefore at higher risk for postoperative pain due to decreased

chest wall compliance. The latter patient was still discharged home earlier than would be expected based on prior reports of patients in his age group. All of our other 7 patients who were >18 years of age (including a 29 year old female) were discharged on POD number 1.

We propose that our addition of bupivacaine intercostal nerve blocks to INC is an important reason that we were able to achieve excellent quality of pain control and maintain consistent hospital lengths of stay. Intercostal nerve cryoablation may take 5 h or more for adequate axonal degeneration and inhibition of nerve impulse transmission to take effect [19], and bupivacaine nerve blocks can help bridge that gap. This delay in onset was well demonstrated in the study by Harbaugh et al., which reported an increased need for intraoperative opioids with cryoablation compared to thoracic epidural (1.08 vs. 0.46 oral morphine equivalents/kg). Moreover, the addition of bupivacaine nerve blocks to INC allowed Rettig et al. to discharge 10 of their most recent 15 patients (66.7%) on the same day of surgery, and they report that none of the patients that required an overnight stay remained in house due to pain. This is a drastic improvement from their recently published series of 40 patients that received cryoablation without bupivacaine nerve blocks, where only 5% of patients were discharged on POD 1 and 60% were discharged on POD 2. Our population of patients seems to consist of patients with more severe pectus deformities than those in the Rettig series, as evidenced by our median Haller index of 4.7 (range 3.1–9.7) compared to that reported in their study (3.6, 3.1–4.6), as well as our inclusion of several adult patients who are known to have significantly increased chest wall stiffness. Nevertheless, we are optimistic that a subset of our patients that are appropriately motivated and operated on early in the day will be able to benefit from same-day discharge as our practices evolve.

The quality of pain control is also notable in our series with a median average postoperative pain score of 2. There was no significant difference in pain scores between patients with one, two or three bars. These pain scores are lower than the average POD 1 pain scores reported in the Graves, et al. series (3.1–3.4) that used cryoablation without bupivacaine nerve blocks [8,9]. Our data are not directly comparable to pain scores reported in the study by Dekonenko et al., who reported daily *maximum* pain scores of 6 and 5 on POD 0 and POD 1, respectively [4]. We attribute our exceptional pain scores to the completeness of our pain control pathway, including our preoperative analgesic regimen (gabapentin, acetaminophen, and celecoxib), our use of intercostal bupivacaine nerve blocks, and cryoablation. In our series, cryoablation included T3-T8 in all cases, whereas all other reports to date are only T3-T7. Importantly, the initiation of gabapentin four days prior to surgery (300 mg TID) permits administration of higher doses postoperatively (400 mg TID).

Narcotics were used sparingly in our series. Our median OME for the inpatient hospital stay was 30 mg (and only 15 mg if you consider patients managed without a PCA), which is less than previously reported. Other series that utilised INC during the Nuss procedure reported a median (Q1-Q3) total postoperative narcotic use of 1.78 (1.26–3.77) mg OME/kg and distribution of a 30–40 dose supply of oxycodone at discharge [10]; 111 mg postoperative OME inpatient and 205 mg OME outpatient [14]; 49.03 mg OME of postoperative IV narcotics alone [11]; and a total postoperative OME of approximately 60 mg [4]. Rettig et al., in reporting same day discharge, did not comment on total opioid usage, but they did note that 67% of their patients took no oxycodone at home [20]. While we were unable to track oxycodone usage at home for those given prescriptions, it is notable that 8/40 patients (20%) took no oxycodone at all, either in the hospital or post discharge, and only three patients requested a refill of their two-day oxycodone prescription, which was routinely given at discharge from the hospital to all patients. Moreover, due to our exceptionally low reported pain scores, we eliminated IVPCA use from our protocols for the most recent 21/40 patients in the series, which resulted in a 73% decrease of total OME with no change in pain scores or discharge timing. After eliminating IVPCA, total postoperative inpatient narcotic intake was equivalent to just two 5 mg oxycodone tablets. Pre-hospital analgesics and upfront use of both intercostal nerve bupivacaine injections and cryoablation prior to bar placement may have the additional benefit of decreasing general anesthetic and opioid requirements intraoperatively. That may have helped in eliminating certain barriers to hospital discharge, such as nausea, somnolence and urinary retention, none of which were present in this series. This is an area of active study for our group.

Return of sensation to the chest wall took longer than expected. The cryoprobe manufacturer describes nerve function returning over a period of 1–3 months, and two other small series have reported return of sensation in 6/10 patients and 18/19 patients by 3 months [9,10]. Another study reported mean time to absence of numbness of 3.4 months (range 1 week to >32 months) in patients ≤21 year old and 10.7 months (range 0 days to >24 months) in older patients [16]. All 36 of our patients who returned for their one month follow-up appointment had a chest wall they invariably described as "still completely numb," and by 3 months postoperatively, only 28% of patients in our series had complete return of sensation. By 6 months postoperatively 85% of our patients had completely normal chest wall sensation, and for the 20 patients who have reached 12 month follow up, all have normal sensation. Of those with some degree of numbness still present at 6 months, the area of numbness was relatively small. None of the patients found these areas of residual numbness bothersome, and further follow up will determine if and when these areas resolve.

In reporting their experience over the first 1–3 months, 4/40 patients (10%) described hypersensitivity to light touch (often noticed when putting on a tee shirt or other light clothing) and 2/40 (5%) described "tingling" sensations. These sensations completely resolved without additional treatment in all patients by 6 months postoperatively and were never reported to be painful in nature. This absence of neuropathic pain with cryoablation is similar to that seen in other studies [16]. Our series comprises the largest and most descriptive report of progression of sensory recovery following INC in patients undergoing a Nuss procedure, and these observations may be useful when counselling patients on what to expect with cryoablation in the postoperative period.

The cryoablation technique is technically straightforward to perform and easily learned, with a short learning curve of just one or two cases. The technique can be learned in an animal cadaver lab or by intraoperative instruction from an experienced surgeon to one new to the technique (as occurred in our group).

Our study does have some limitations. Analysis is limited by our sample size, even though our series is one of the largest reported. Moreover, while we have detailed information on in-hospital opioid consumption, we were unable to track detailed data regarding oxycodone consumption following discharge and we did not query patients on their subjective impressions of post-operative pain. There may have been patients who took little or no oxycodone that were not identified, and this is an area to improve in future studies. The fact that our clinical protocol minimizes narcotic usage biases narcotic usage as an outcome measure. Decisions regarding readiness for discharge affect LOS making it a subjective variable. Another limitation of our study with respect to determining return of sensation is that there is a gap in our follow-up intervals. Since after the 3 month visit patients were not seen again until 6 months, we cannot be specific on timing for return of

sensation in those patients with recovery after 3 months but before 6 months or after 6 months and before 12 months. We are considering some type of additional follow up between 3 and 6 months to better clarify the return of nerve function for future patients.

5. Conclusion

In summary, intercostal nerve cryoablation combined with intercostal bupivacaine nerve blocks and a well-defined pre- and post-hospital enhanced recovery after surgery (ERAS) analgesic protocol was successful in discharging patients one day after the Nuss procedure, eliminating the need for intravenous opioids and achieving excellent pain control. Return of nerve function took longer than expected, but 85% of patients that reached 6 month follow-up and 100% of patients that reached 12 month follow-up had normal sensation with no areas of residual numbness. Importantly, there were no cases of chronic neuropathic pain.

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